Signal Denoising
Report 1

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1 Introduction

This study presents a personal assessment of two signal denoising approaches applied to a synthesized ECG trace. The study compares the performance of a linear filtering method (low-pass Butterworth filter) and a non-linear denoising method based on FFT thresholding. In this work, the ECG signal is synthesized using the `ecgsyn` function. The noise is added to obtain an SNR of 1 (0 dB), and then the two denoising methods are applied. For each method, the performance metrics used are: relative RMS bias, SNR improvement, and Maximum Absolute Error (MAE).

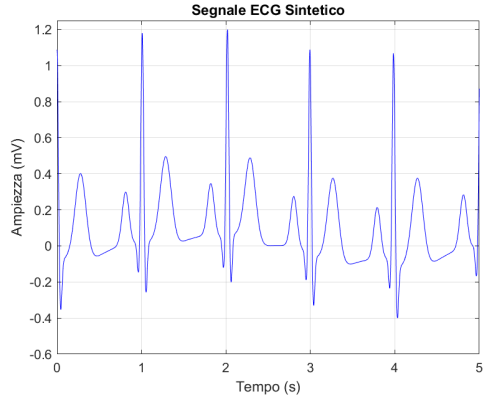
2 Parameters

2.1 Simulation Parameters

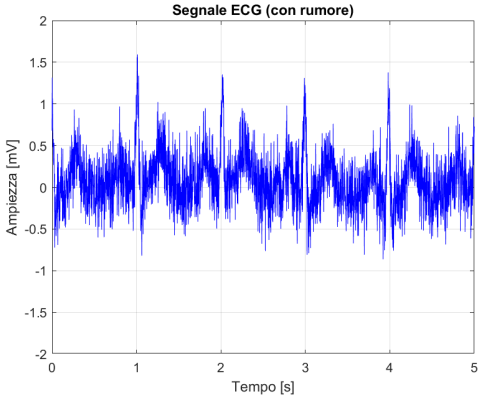
The synthesized ECG signal is generated with the `ecgsyn` function with this parameters:

- **Sampling frequency,** `sfecg` = 512 Hz
- **Internal frequency,** `sfint` = 2×`sfecg`
- **Number of beats,** `Nbeats` = 10, which provides a length of approximately 15 seconds. In this report, the plot will have a duration of 5 seconds.
- **ECG parameters:** mean heart rate = 60 bpm, standard deviation = 1 bpm, LF/HF ratio = 0.5.

The standard deviation of the noise is equal to the RMS of the clean ECG signal. This provides that the signal and noise power are equal (SNR = 1, 0 dB).



2.1 Original signal



2.2 Added noise to previous signal

2.2 Filter-Based Denoising

For the filter-based approach, is used the Butterworth filter considering the following parameters:

- **Filter passband, BP** = 40 Hz.
- **Filter stopband, BS** = 1.1, 2, 5 × BP.

- **Attenuation, A** = 100, 1000.
- **Passband ripple, R** = 0.01 (1%).

The stopband frequency is normalized following the Nyquist frequency (which is `sfecg/2`). The Butterworth filter is designed using `buttord` and `butter`, with a maximum order of 8. When the Butterworth filter is applied to an ECG signal, it generates some delay. In order to avoid this, the function `filtfilt` is used, which eliminates the delay.

2.3 Non-Linear Denoising

In this approach, the noisy signal is first centered by subtracting its mean. A Hann window is applied to reduce spectral leakage. Then, after calculating the FFT of the signal, its magnitude is normalized by dividing by the sum of the window values. A threshold is set to select only those FFT coefficients whose magnitude is greater than a chosen value. The threshold is defined by the formula:

$$th = k \times \text{noise_std} \times \sqrt{Nsample}$$

with $k = 1, 3, 10$. Coefficients below the threshold are set to zero. The denoised signal is reconstructed using the inverse FFT (considering only the real part).

3 Evaluation Metrics

The performance of the two denoising methods was evaluated using the following metrics:

- **Relative RMS Bias:** measures the systematic error between a reconstructed signal and a reference. It is the squared mean difference (bias) normalized by the signal's power:

$$\frac{\text{RMS}_{\text{bias}}^2}{\text{RMS}_s^2} \tag{1}$$

- **SNR Improvement:** the improvement in decibels (dB) is computed as the difference between the SNR of the denoised signal and the SNR of the noisy signal:

$$\frac{SNR_{out}}{SNR_{in}} = 10\log_{10}\left(\frac{\sum_{n=1}^{N} s(n)^2}{\sum_{n=1}^{N}(s(n)-\hat{s}(n))^2}\right) - 10\log_{10}\left(\frac{\sum_{n=1}^{N} s(n)^2}{\sum_{n=1}^{N}(s(n)-s_{\text{noisy}}(n))^2}\right) \tag{2}$$

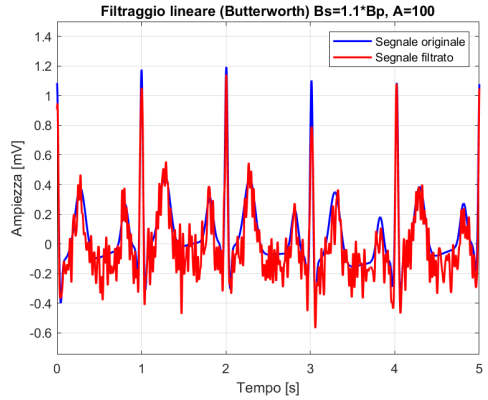
- **Maximum Absolute Error (MAE):** this is the maximum absolute difference between the original signal and the denoised signal, defined as:

$$\max|s(n)-\hat{s}(n)| \tag{3}$$

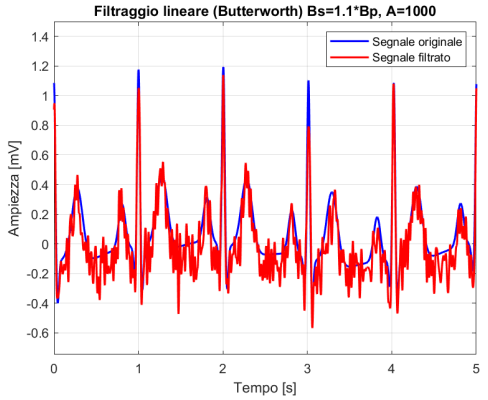
4 Results

In this section are showed the plots of the signals and the tables with the results calculated by the evaluation metrics.

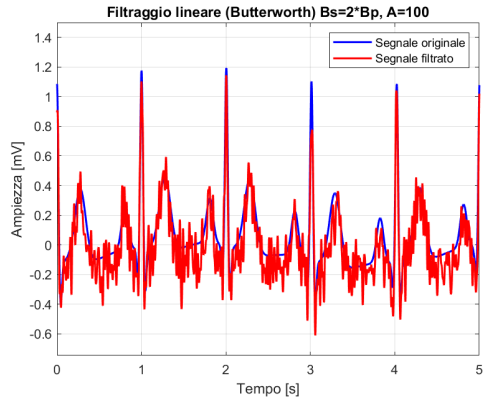
4.1 Filter-Based Denoising



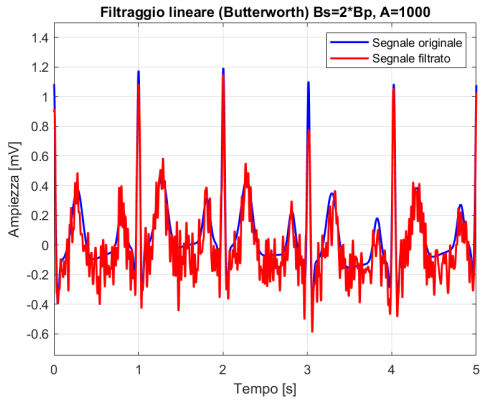
4.1 Bs=1.1×Bp, A=100



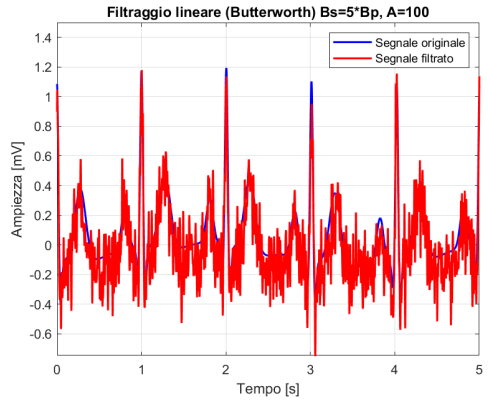
4.2 Bs=1.1×Bp, A=1000



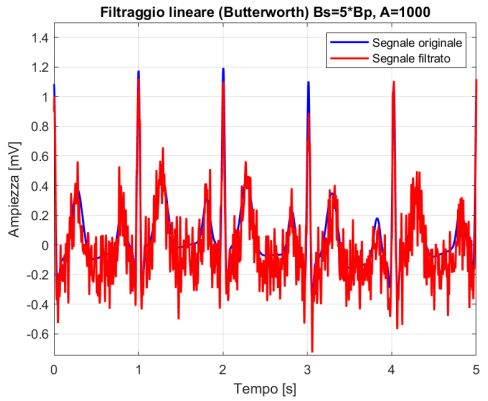
4.3 Bs=2×Bp, A=100



4.4 Bs=2×Bp, A=1000



4.5 Bs=5×Bp, A=100



4.6 Bs=5×Bp, A=1000

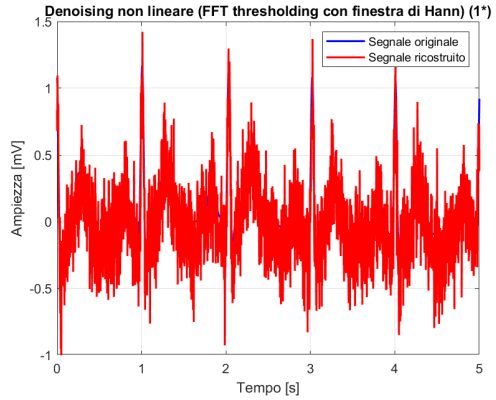
Figure 4.1: Results from the Filter-based denoising

The results obtained for the filter-based method (Butterworth filter) with various filter specifications were:

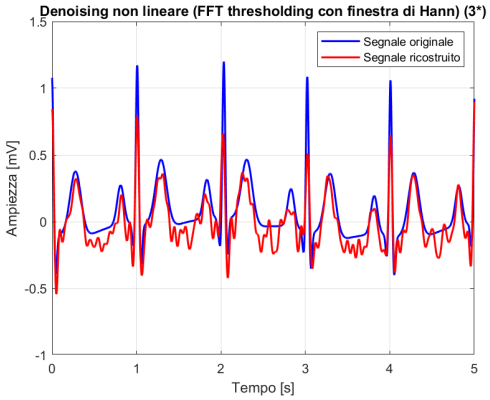
Butterworth filter	**relative RMS Bias**	**SNR Improvement (dB)**	**MAE**
BS = 1.1×BP, A = 100	0.1558	5.33	0.4258
BS = 1.1×BP, A = 1000	0.1714	5.34	0.4520
BS = 2×BP, A = 100	0.1415	5.53	0.4310
BS = 2×BP, A = 1000	0.1924	4.61	0.4798
BS = 5×BP, A = 100	0.1604	3.94	0.5865
BS = 5×BP, A = 1000	0.1082	3.98	0.5650

Table 1: Results for filter-based denoising with different filter stopband and attenuation parameters.

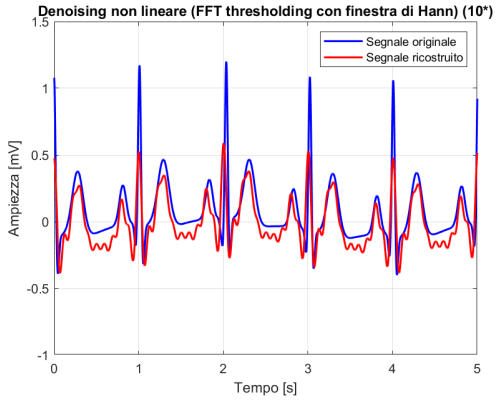
4.2 Non-Linear Denoising



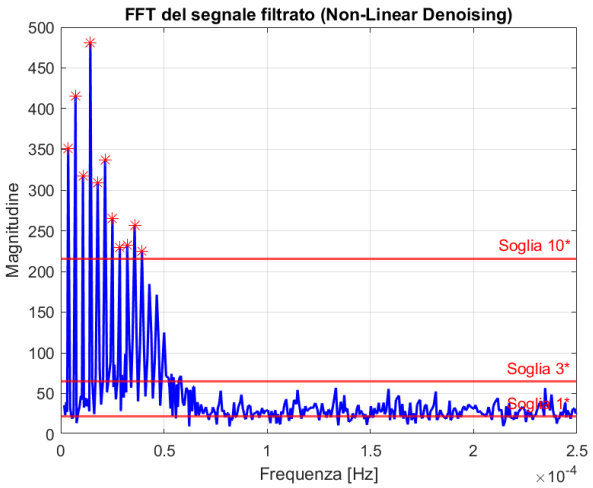
4.7 th=$1\times\sigma_n\times\sqrt{N}$



4.8 th=$3\times\sigma_n\times\sqrt{N}$



4.9 th=$10\times\sigma_n\times\sqrt{N}$



4.10 Frequencies domain of the best case ($3\times\sigma_n\times\sqrt{N}$) with three thresholds

For the non-linear FFT thresholding method, the following results were obtained with different threshold multipliers:

Threshold Multiplier	**relative RMS Bias**	**SNR Improvement (dB)**	**MAE**
$1\times\sigma_n\times\sqrt{N}$	0.1804	0.35	0.9406
$3\times\sigma_n\times\sqrt{N}$	0.1154	5.95	0.5603
$10\times\sigma_n\times\sqrt{N}$	0.1648	4.62	0.7143

Table 2: Results for non-linear FFT thresholding denoising using different threshold multipliers.

5 Discussion and Conclusion

The analysis shows that using a Butterworth filter for noise removal can work well if the filter settings are chosen carefully. In this tests, the best result was achieved with Bs set to 2 times Bp and an attenuation of 100. This gave an SNR improvement of about 5.5 dB and a low MAE. The peaks are almost preserved and the signal is better reconstructed than the other cases. While this case provides a good balance between noise removal and signal preservation, the other cases show an imbalance between these two aspects. For example, when Bs is 1.1×Bp and A=1000, the attenuation is too high, which reduces noise but also removes important signal components, losing useful information. Another example, when Bs is 5×Bp and A=100, the figure shows the accuracy of the denoised signal is reduced, this happens because the stopband is too wide, so many useful frequencies are lost, as confirmed by the table. The MAE is the highest in this case.

The non-linear FFT thresholding method also gave good results. When using a threshold multiplier of 3, it had a better RMS bias than the Butterworth filter and improved the SNR by almost 6 dB. Focusing on the peaks of 3× and 10×, it's possibile to observe their attenuation. This happens because noise filtering limits the amplitude of the signal. The performance of this method changes with different multipliers. A lower value (1×) does not remove enough noise, while a higher value (10×) removes too much useful signal.

Hence, a multiplier of 3 offers a trade off between lower noise and higher preserved signal. As a proof, in Fig.4.10 it's possible to focus on the frequency domain. The frequency components above the "1×" threshold let through many noise components and don't allow proper signal reconstruction. The other thresholds behave differently: they are able to discard the noise components but may also lose some peaks. In particular, the 10× threshold follows the trend of the signal but loses amplitude.

In conclusion, both methods have strengths and weaknesses. The Butterworth filter gives acceptable but still improvable results, while the FFT method is more sensitive to the chosen multiplier threshold.